

CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

# Chapter 20: Area and Perimeter of Plane Figures

Every formula and result of the chapter on one sheet

|    |                          |                                                                |
|----|--------------------------|----------------------------------------------------------------|
| 1  | Semi-perimeter           | $s = \frac{a + b + c}{2}$                                      |
| 2  | Heron's formula          | $\sqrt{s(s-a)(s-b)(s-c)}$                                      |
| 3  | Equilateral triangle     | Area $\frac{\sqrt{3}}{4}a^2$ , height $\frac{\sqrt{3}}{2}a$    |
| 4  | Isosceles triangle       | $\frac{b}{4}\sqrt{4a^2 - b^2}$                                 |
| 5  | Rectangle                | Area $lb$ , perimeter $2(l + b)$ , diagonal $\sqrt{l^2 + b^2}$ |
| 6  | Square                   | Area $a^2 = \frac{1}{2}d^2$ , perimeter $4a$                   |
| 7  | Parallelogram            | Base $\times$ height                                           |
| 8  | Rhombus                  | $\frac{1}{2}d_1d_2$                                            |
| 9  | Trapezium                | $\frac{1}{2}(a + b)h$                                          |
| 10 | Circle                   | Circumference $2\pi r$ , area $\pi r^2$                        |
| 11 | Ring                     | $\pi(R^2 - r^2)$                                               |
| 12 | Semicircle               | Area $\frac{1}{2}\pi r^2$ , perimeter $\pi r + 2r$             |
| 13 | Path outside a rectangle | $(l + 2w)(b + 2w) - lb$                                        |
| 14 | Cross-paths              | $w(l + b - w)$                                                 |